

Information Security

Computing and Mathematics

May, 2026



Short Title:	Information Security	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Forensics and Security	
Author	JMCGIBNEY	
Credits	10	Level: Postgraduate

Description of Module / Aims

This module provides students with a solid grounding in the area of computer and information security. The question of securing modern distributed applications is treated holistically, where students assess security threats and vulnerabilities at different layers and identify suitable services and technologies to help address these threats. Cryptographic foundations that underpin many security mechanisms are covered in some detail, as well as best practices in secure software development. Students will also discuss incident handling and a selection of techniques and tools for forensic investigations.

Programmes

COMP-0973	MSc in Computer Science (Enterprise Software Systems) (WD_KCESS_R)	1 / 0 / E
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Indicative Content

- Security threats: attack methods; common vulnerabilities; threat modelling; security services
- Cryptography: symmetric block and stream ciphers; public key cryptography; hash functions; authentication schemes; key management & certificates
- Network and cloud security: TLS; VPNs; wireless security; identity and access management; firewalls; intrusion prevention
- Software security: secure software development practices; OWASP Top 10; malware
- Security assessment: black and white box testing; penetration testing; standards, compliance
- Incident response & recovery: steps involved in incident response; the forensic process; data recovery; computer forensics

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate the various security threats and attacks to which an organisation may be susceptible.
2. Compare and contrast leading cryptographic techniques and determine their applicability in a variety of practical scenarios.
3. Assess key aspects of system, network and cloud security in a heterogeneous computing environment.
4. Evaluate the specific security concerns pertinent when developing and deploying software applications.
5. Carry out a security assessment on a modern distributed application.
6. Recommend incident response approaches and assemble digital evidence from a variety of sources.

Learning and Teaching Methods

- Combination of lectures and guided computer-based practical exercises.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,6
Continuous Assessment	50%	1,2,3,4,5,6
<i>Assignment</i>	20%	1,2,3
<i>Assignment</i>	20%	4,5
<i>Assignment</i>	10%	6

Assessment Criteria

- <40%:** Unable to interpret and describe key concepts of the specific knowledge domains of computer security.
- 40%-59%:** Able to interpret, describe and discuss key concepts of the specific knowledge domains of computer security. Able to discover and integrate related knowledge to assess and implement security approaches.
- 60%-69%:** Able to solve problems within the specific knowledge domains by experimenting with the appropriate skills and tools.
- 70%-100%:** All the above to an excellent level. Be able to propose solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	24
Practical	24	24
Independent Learning	222	222

Essential Material

- "Open Web Application Security Project (OWASP)." <https://www.owasp.org/>
- "The SANS Institute." <https://www.sans.org/>

Supplementary Material

- McGraw, G. _Software Security: Building Security In_. Boston: Addison-Wesley, 2006.
- Stallings, W. and L. Brown. _Computer Security: Principles and Practice_. 4th ed.. Harlow: Pearson, 2018.